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Weekly report 25: Preeclampsia Detection through Fetal Ultrasound images with Artificial Intelligence

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1 Progress

1.1 Preeclampsia classification based on placental segmentation

The results for classification based on placental segmentations are shown in Table 1.

Data/Model	CNN	ResNet18
Image	62.01%	71.54%
Masked image	50.00%	61.40%
Masked placenta	62.13%	70.40%
Masked image, image brightness	53.70%	59.90%
Only mask	54.57%	58.17%

Table 1: Results for the accuracy on the validation set for different types of data regarding placental images. An example of what each row entails is shown in Figure 5

Grad-CAM has been calculated on the same set of 50 images from the test set for both models, CNN and ResNet18. Some examples of the overlayed heatmaps are shown in Figure 10. The conclusions extracted from model experiments and Grad-CAM are:

- Introducing the whole image to the models is the input that holds the biggest predictive power.
- Placenta is not as important by itself. The whole context is more valuable.
- Average image pixel brightness does not bring extra information.
- Mask shape or position are not important features for the models' decision.
- CNN model falls into shortcut learning, while ResNet18 model is more focused on the anatomy. Hence, CNN results are not reliable.

Preprocessed image

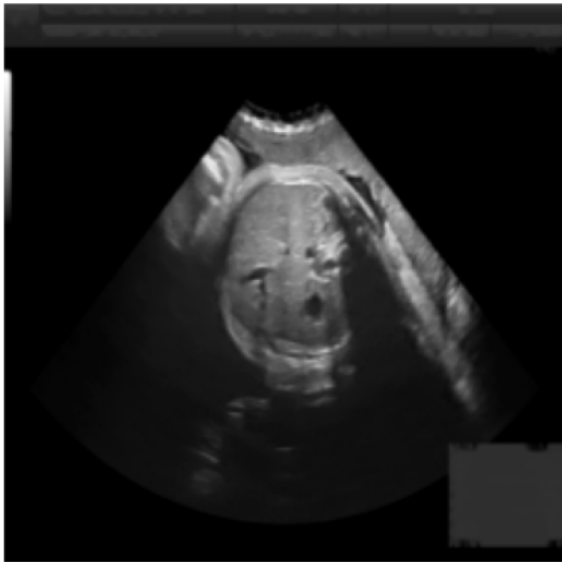


Figure 1: Processed image.

Placental mask



Figure 2: Mask.

Maked image



Figure 3: Masked image.

Masked placenta

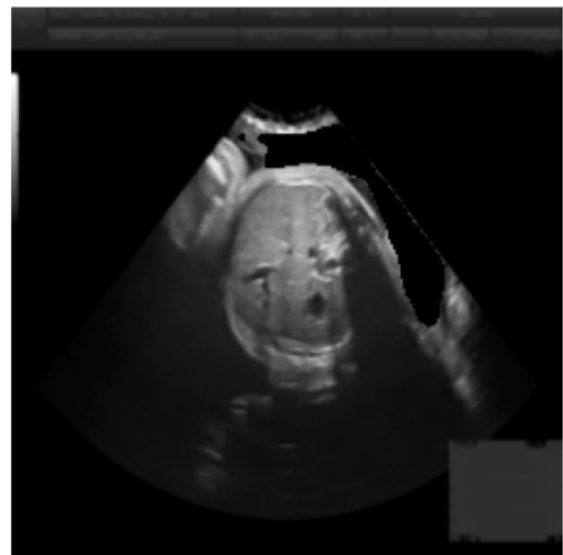


Figure 4: Masked placenta

Figure 5: Example of a preprocessed image, its placental mask, and other combinations.

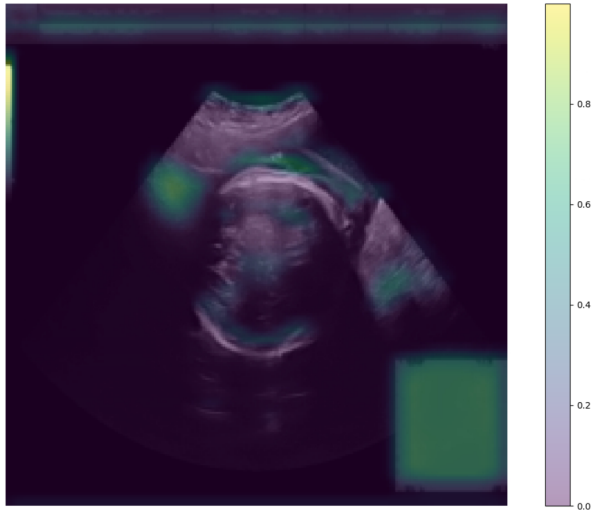


Figure 6: Head plane with Grad-CAM on CNN.

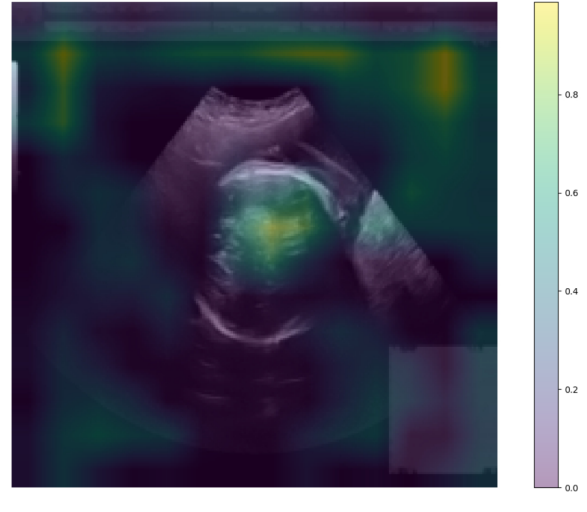


Figure 7: Head plane with Grad-CAM on ResNet18.

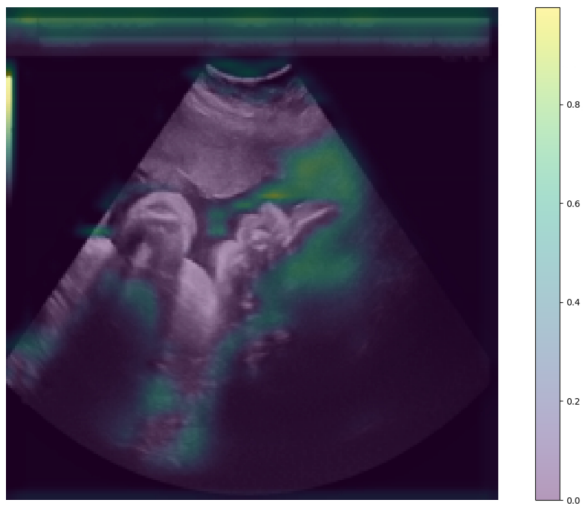


Figure 8: Profile plane with Grad-CAM on CNN.

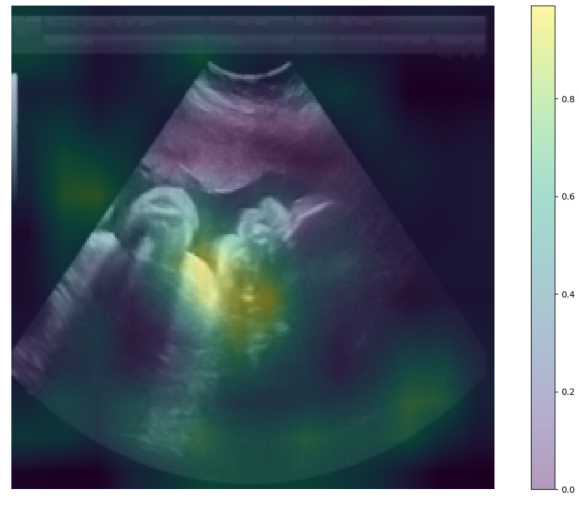


Figure 9: Profile plane with Grad-CAM on Resnet18.

Figure 10: Grad-CAM overlayed over head and profile planes from CNN and ResNet18 for comparison.

References